Reg. No. | | | | | | | | | | | [14CS/IS32]

THIRD SEMESTER B.TECH EXAMINATION DECEMBER –2015

DISCRETE MATHEMATICAL STRUCTURES AND GRAPH THEORY

Time: 3 Hours **Maximum Marks: 70**

Instruction: Answer any **FIVE** full questions, choosing at least **TWO** from each part.

PART-A

1. a. A survey of 500 television viewers of a sports channel produced the following information: 285 watch cricket, 195 watch hockey, 115 watch football, 45 watch cricket and football, 70 watch cricket and hockey, 50 watch hockey and football and 50 do not watch any of the three kinds of games. **08**

a. How many viewers in the survey watch all three kinds of games?

b. How many viewers watch exactly one of the sports?

b. For any two sets A and B, prove the DeMorgan's Laws: **6**

i. $\overline{A \cup B} = \overline{A} \cap \overline{B}$

ii. $\overline{A \cap B} = \overline{A} \cup \overline{B}$

2. a. Prove that, for any propositions p, q, r the compound proposition **06**

$$\left[(p \to q) \wedge (q \to r)\right] \to (p \to r)$$

b. Test whether the following is a valid argument: **04**

If I study, then I do not fail in the examination. If I do not fail in the examination, my father gifts a two-wheeler to me.

∴ If I study then my father gifts a two wheeler to me.

c. Give a direct proof of the statement: **04**

" The square of an odd integer is an odd integer"

3. a. Prove that if 30 dictionaries in a library contain a total of 61, 327 pages, then at least one of the dictionaries must have at-least 2045 pages. **03**

b. Prove the statement: If m=Kn+1 pigeons (where K≥1) occupy n pigeonholes then atleast one pigeonhole must contain K+1 or more pigeons. **03**

c. Find all positive integers n (≥2) for which for which the complete graph K_n contains an Euler circuit. For which n does K_n have an Euler trail but not an Euler circuit? **04**

d. Show that every simple K-regular graph with 2k-1 vertices is a Hamilton graph. **04**

4. a. Show that A graph of order 5 and size 8 and A graph of order 6 and size 12 are planar graphs. **05**

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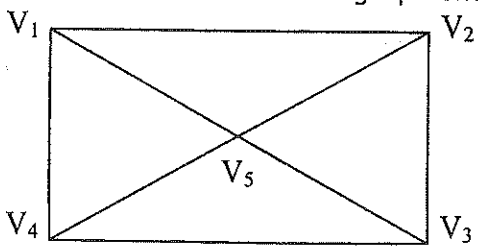
b. A connected planner graph G and 9 vertices with degrees 2,2,3,3,3,4,5,6,6. Find the number of regions of G. **05**

c. Find the chromatic number of the graph shown below: **04**



PART-B

5. a. Apply Warshall's algorithm to compute the transitive closure (path matrix) of the digraph defined by the following adjacency matrix. **10**

Fig

	1	2	3	4
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	0	0	0	0

b. Let **04**

$A = \{1, 2, 3, 4\}$, *and*

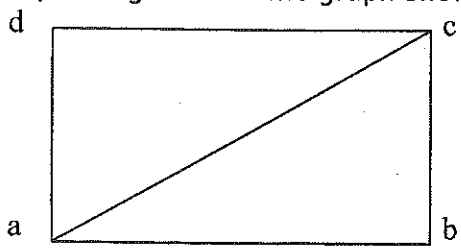
$R = \{(1,1), (1,2), (2,2), (2,4), (1,3), (3,3), (3,4), (1,4), (4,4)\}$

Verify that R is a partial order on A. Also write down the Hasse diagram for R.

6. a. "Any connected graph with n vertices and n-1 edges is a tree"Prove the theorem. **04**

b. A class room contains 25 microprocessors that must be connected to a wall socket that has 4 outlets. Connections are made by using extension cards that have 4 outlets each. Find the least number of cards needed to get this computer set up for the class. **04**

c. Draw all the spanning trees of the graph shown below: **06**



7. a. Prove the following theorem: **06**

Let G can be a group and let a,b,x be elements of G then

i. xa = xb implies a=b

ii. ax = bx implies a=b

b. Define **08**

i. Semigroups

ii. Monoids

iii. Homomorphism of semigroup

iv. Isomorphism

8. **a.** Prove that if $A = \{a_1, a_2 ... a_n\}$ be a finite set with n elements, n≥2, then there are n!/2 even permutation and n!2 odd permutation. **10**

b. Show that f(n) = $2n^3+9n+5$ is $O(n^2)$ **04**